

3. Introduction to four vectors

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1 Introduction to four vectors

We have studied the transformation.

$$\Delta x' = (\Delta x - u \Delta t) \gamma$$

$$\Delta t' = \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma$$

Frame S

$$\begin{cases} E_1 \text{ occurs at } (x_1, t_1) \\ E_2 \text{ occurs at } (x_2, t_2) \end{cases} \quad \text{If } (t_2 > t_1) \Rightarrow \Delta t = t_2 - t_1 > 0$$

We can have frame S' moving with speed u such that

$$\Delta t' < 0 \quad \text{i.e.} \quad E_2 \text{ occurs earlier than } E_1$$

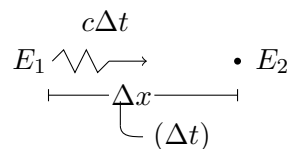
$$\Delta t - \frac{u \Delta x}{c^2} < 0$$

$$c \Delta t < \frac{u}{c} \Delta x$$

$$\boxed{\frac{u}{c} > \frac{c \Delta t}{\Delta x}}$$

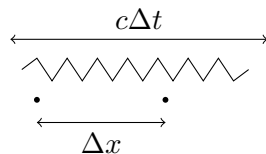
u is always less than $c \quad \therefore \frac{u}{c} < 1$

$$\Rightarrow c \Delta t < \Delta x \quad (\text{space like events})$$



If Δx is so large that even $c \Delta t < \Delta x$ than there exist an inertial frame in which order of events can be reversed

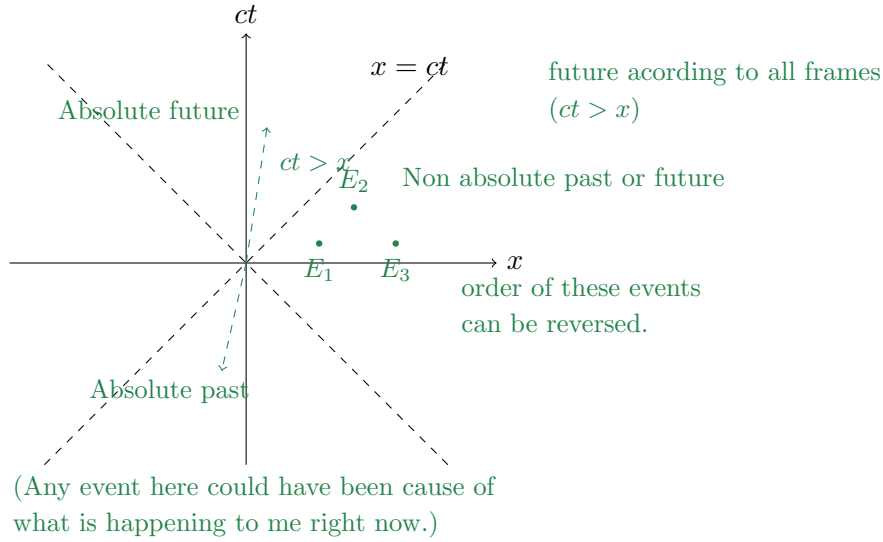
If we have $\Delta x < c \Delta t$



Then $\frac{u}{c} > 1$ which is not possible.

and we can't reverse order of event. Such events can be causally connected and therefore order of events can not be reversed.

(If there is enough time for light to reach Δx distance then these two events can be causally connected).



Four vector :

It has 4 components

$$x = (x_0, x_1, x_2, x_3)$$

where $(x_0 = ct)$ & $\beta = u/c$

$$\text{and } \left. \begin{array}{l} x_1 = x \\ x_2 = y \\ x_3 = z \end{array} \right\} \text{ for our 3D world}$$

$$x = (x_0, \vec{r})$$

In terms of four vectors Lorentz transformation looks like

$$x' = (x - ut) \gamma$$

$$x'_1 = \left(x_1 - \left(\frac{u}{c} \right) \cdot ct \right) \gamma = (x_1 - \beta x_0) \gamma$$

and,

$$t' = \left(t - \frac{ux}{c^2} \right) \gamma$$

$$ct' = \left(ct - \frac{u}{c} \cdot x \right) \gamma$$

$$x'_0 = (x_0 - \beta x_1) \gamma$$

So Transformation becomes :-

$$x'_0 = (x_0 - \beta x_1) \gamma$$

$$x'_1 = (x_1 - \beta x_0) \gamma$$

$$x'_2 = x_2$$

$$x'_3 = x_3$$

$$\text{or} \quad \bar{x}_i = \Lambda_j^i x_j$$

$$\text{where} \quad \Lambda_j^i = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2 The structure of space time :-

(i) Four vectors :-

The lorentz Transformation takes on simple appearance if we express them in terms of quantities.

$$x^0 = ct, \quad \beta = u/c$$

(1 m of x^0 = time it takes to cover 1 m using speed c)

We express,

$$\begin{aligned} x &= x^1 \\ y &= x^2 \\ z &= x^3 \end{aligned}$$

Then lorentz transformation reads

$$\left. \begin{aligned} \bar{x}^0 &= (x^0 - \beta x^1) \gamma \\ \bar{x}^1 &= (x^1 - \beta x^0) \gamma \\ \bar{x}^2 &= x^2 \\ \bar{x}^3 &= x^3 \end{aligned} \right\} \quad \text{or} \quad \boxed{\begin{pmatrix} \bar{x}^0 \\ \bar{x}^1 \\ \bar{x}^2 \\ \bar{x}^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}}$$

$$\text{or} \quad \bar{x}^\mu = \sum_{\nu=0}^3 (\Lambda^\mu_\nu) x^\nu$$

$$\text{or} \quad \bar{x}^i = \sum_{j=0}^3 (\Lambda^i_j) x^j$$

Where ‘ Λ ’ is lorentz transformation matrix.

Superscript ‘ μ ’ labels row and subscript ‘ ν ’ labels column.

We are interested in change of components when you go to a moving system.

3 vector is defined as any set of 3 components that transform under rotation the same way (x, y, z) do;

By extension we now define 4 vector as Any set of 4 components that transform in same manner as (x^0, x^1, x^2, x^3) under lorentz transformations

$$\bar{a}^\mu = \sum_{\nu=0}^3 (\Lambda^\mu_\nu) a^\nu = \Lambda^\mu_\nu a^\nu$$

Where a^ν is a four vector.

For perticular case of transformation along x-axis

$$\begin{aligned} \bar{a}^0 &= (a^0 - \beta a^1) \gamma \\ \bar{a}^1 &= (a^1 - \beta a^0) \gamma \\ \bar{a}^2 &= a^2 \\ \bar{a}^3 &= a^3 \end{aligned}$$

Four vector analog of dot product $(\vec{A} \cdot \vec{B} = \sum_{i=1}^3 A_i B_i = A_i B_i)$

But it is defined with zeroth component with minus sign.

$$a_\mu b^\mu = a_0 b^0 + a_1 b^1 + a_2 b^2 + a_3 b^3$$

Here covariant vector (a_μ) differs from contravariant (a^μ) only in sign of zeroth component.

$$a_\mu = (a_0, a_1, a_2, a_3) \equiv (-a^0, a^1, a^2, a^3)$$

Raising or lowering the temporal index costs a minus sign.

$$(a_0 = -a^0) \quad \text{or} \quad (a^0 = -a_0)$$

But lowering or raising a spatial index changes nothing

$$(a^1 = a_1), (a^2 = a_2), (a^3 = a_3)$$

So our dot product of two four vectors become

$$a_\mu b^\mu = a_0 b^0 + a_1 b^1 + a_2 b^2 + a_3 b^3$$

$$a_\mu b^\mu = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3$$

This is 4 dim scalar product and just like dot product $(A_i B_i)$ is rotational invariant, This 4 vector dot product is invariant under lorentz transformations.

$$a_\mu b^\mu = \bar{a}_\mu \bar{b}^\mu$$

$$\text{i.e.} \quad -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3 = -\bar{a}^0 \bar{b}^0 + \bar{a}^1 \bar{b}^1 + \bar{a}^2 \bar{b}^2 + \bar{a}^3 \bar{b}^3$$

Where

$$a_\mu = \sum_{\nu=0}^3 g_{\mu\nu} a^\nu \quad g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\left. \begin{array}{l} a_0 = -a^0 \\ a_1 = a^1 \\ a_2 = a^2 \\ a_3 = a^3 \end{array} \right\} \quad (\text{Minkowski metric})$$

Scalar product can be written as $\sum_{\mu=0}^3 a^\mu b_\mu$ or $\sum_{\mu=0}^3 a_\mu b^\mu$

or more compactly as

$$a^\mu b_\mu \quad \text{or} \quad a_\mu b^\mu$$

(Summation is implied whenever a greek index is repeated in a product once as a covariant index and once as a contravariant)

$$\left. \begin{array}{l} \text{So} \quad \sum_{\nu=0}^n a_\nu b^\nu = a_\nu b^\nu = a^\nu b_\nu \\ \text{But} \quad \sum_{\nu=0}^n a_\nu b_\nu \neq a_\nu b_\nu \end{array} \right\} \quad \text{Is it so?}$$

We can write these as

$$\begin{aligned} a_\mu b^\mu &= a^\mu b_\mu = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3 \\ &= -a_0 b_0 + a_1 b_1 + a_2 b_2 + a_3 b_3 \end{aligned}$$

(ii) The Invariant Interval

We know that any vector in 4D minkowski space has four components

$$a^\mu = (a^0, a^1, a^2, a^3)$$

If we go to another frame these components transform under Lorentz transformation

$$\bar{a}^\mu = \sum_{\nu=0}^3 (\Lambda^\mu_\nu) a^\nu = \Lambda^\mu_\nu a^\nu$$

$$\text{length/norm of a 4-vector} = a^\mu a_\mu = a_\mu a^\mu = -(a^0)^2 + (a^1)^2 + (a^2)^2 + (a^3)^2$$

$$|a| > 0 \quad \text{if spatial terms beat temporal term (space like)}$$

$$\text{length of 4 vector} \quad |a| = 0 \quad \text{light like}$$

$$|a| < 0 \quad \text{Time like} \quad (\text{temporal term dominate})$$

Displacement 4-vector :-

Suppose event A occurs at $(x_A^0, x_A^1, x_A^2, x_A^3)$ & event B occurs at $(x_B^0, x_B^1, x_B^2, x_B^3)$

Then the diff $\Delta x^\mu = x_A^\mu - x_B^\mu$ is called displacement 4-vector.

The dot product of Δx^μ with itself $\Delta x^\mu \Delta x_\mu =$ Invariant Intervals b/w two events. (I)

$$\underline{I} = \Delta x^\mu \Delta x_\mu$$

$$\text{Where,} \quad \Delta x^\mu = x_A^\mu - x_B^\mu = \begin{pmatrix} x_A^0 - x_B^0 \\ x_A^1 - x_B^1 \\ x_A^2 - x_B^2 \\ x_A^3 - x_B^3 \end{pmatrix}$$

$$\begin{aligned} \Delta x_\mu \Delta x^\mu = I &= -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 \\ &= -(c\Delta t)^2 + (\Delta x)^2 \end{aligned}$$

‘ Δt ’ is time diff b/w two events & Δx is spatial diff.

When we transform Δx^μ to another frame of reference, Δt & Δx both changes to $\Delta t'$ & $\Delta x'$ but Interval I ($\Delta x^\mu \Delta x_\mu$) remains same.

If i) $I < 0$ i.e. $c\Delta t > \Delta x$ (order of events can't be reversed)

$\exists$ a frame where $\Delta x' = 0$ (these events occur at same place)

$$\Delta x' = (\Delta x - u \Delta t) \gamma$$

$$\text{for } \Delta x' = 0 \quad \left(u = \frac{\Delta x}{\Delta t} \right)$$

$$\text{If } u = \frac{\Delta x}{\Delta t} \quad \text{we have such frame.}$$

ii) If $I > 0$ i.e. $\Delta x > c\Delta t$ (order of events can be reversed)

$\exists$ a frame where both events occur simultaneously

$$\Delta t' = \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma$$

$$\text{for } c\Delta t = \frac{u}{c} \Delta x$$

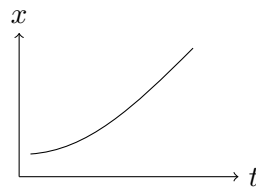
$$\frac{u}{c} = \beta = \frac{c\Delta t}{\Delta x} < 1 \quad \text{allowed.}$$

$$u = c \left(\frac{c\Delta t}{\Delta x} \right)$$

iii) $I = 0$ $c\Delta t = \Delta x$ then two events could be connected by a light signal.

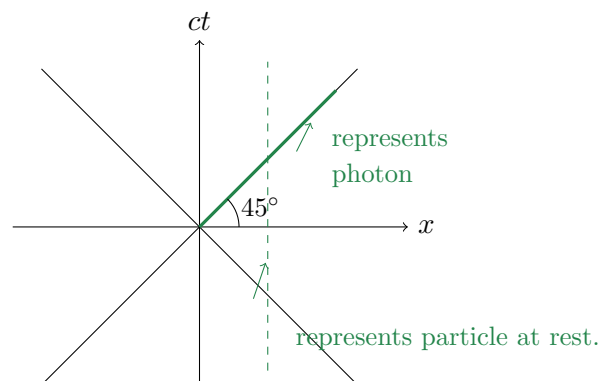
(iii) Space time diagrams :-

Generally we plot



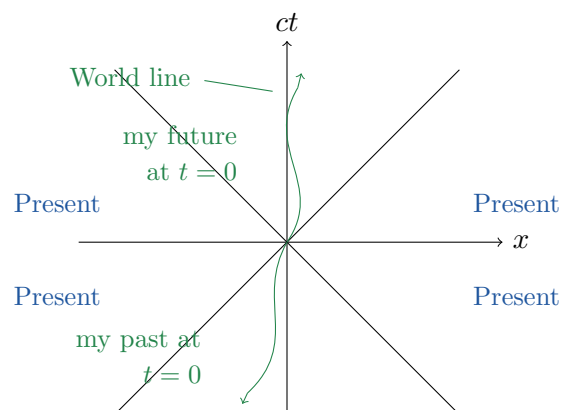
where slope is instantaneous velocity of particle

For some reason convention is reversed, and we plot time on vertical axis and position on horizontal axis. here velocity is given by reciprocal of slope.



$$\text{Slope} = \frac{c\Delta t}{\Delta x} \quad \text{Slope} = \frac{c}{v} = \frac{1}{\beta}$$

Plot on such diagrams are called as “minkowski diagrams”



Any trajectory of a particle on minkowski diagram is called a world line.

Past :– Locus of all points from which I might have come.

Future :– It is locus of all points accessible to me

Present :– Locus of all inaccessible points (i.e. I can't get there, and I did not came from there)
i.e. There is no way I can Influence any event of present.

$$\begin{aligned} \text{for slope} \left(= \frac{c\Delta t}{\Delta x} \right) &> 1 && : \text{time like events} \\ \text{slope} &< 1 && : \text{space like event} \\ \text{slope} &= 1 && : \text{light like event.} \end{aligned}$$

We can also draw y axis as coming out of paper, then we have light cones as boundaries. where forward light cone is future at $t = 0$, backward light cone represents past.

If we can draw z axis, boundaries become hypercones.

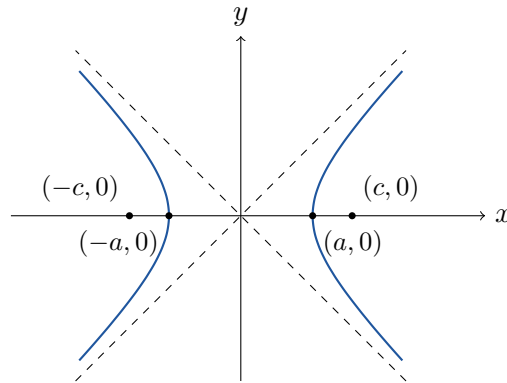
Hyperbolic geometry of space time

Space & time are thought to be a union but remember that time is nothing like space, The distinction arise in minus sign of space time interval

$$I = (\Delta x)^\mu (\Delta x)_\mu = -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2$$

This minus sign imparts space time a hyperbolic geometry.

Just like under rotation about z axis, $\sqrt{x^2 + y^2}$ is preserved, whose locus of all points with fixed $\sqrt{x^2 + y^2}$ represents circle. Under Lorentz transformations, the interval $I = c^2 t^2 - x^2$ is preserved. and locus of all points with given value of I represents hyperbola. which is more rich than circular geometry.



[At this point in the notebook a printed reference figure and text on the standard form of the equation of a hyperbola were pasted in from a textbook; the sketch above is a redrawn stand-in and the textbook prose is not reproduced here.]

$$\begin{aligned} \bar{x}^0 &= (x^0 - \beta x^1) \gamma \\ \bar{x}^1 &= (x^1 - \beta x^0) \gamma \\ \bar{x}^2 &= x^2 \\ \bar{x}^3 &= x^3 \end{aligned} \quad \begin{aligned} x^\mu x_\mu &= -(x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2 \\ &\uparrow \\ &\text{space time interval b/w origin and point } (x, t). \end{aligned}$$

$$\begin{aligned}
& -(\bar{x}^0)^2 + (\bar{x}^1)^2 + (\bar{x}^2)^2 + (\bar{x}^3)^2 \\
& = -\gamma^2 ((x^0)^2 + \beta^2(x^1)^2 - 2\beta x^0 x^1) + \gamma^2 ((x^1)^2 + \beta^2(x^0)^2 - 2\beta x^1 x^0) + (x^2)^2 + (x^3)^2 \\
& = \gamma^2 ((x^1)^2 - (x^0)^2 + \beta^2 ((x^0)^2 - (x^1)^2)) + (x^2)^2 + (x^3)^2 \\
& = \gamma^2 ((x^1)^2 - (x^0)^2) (1 - \beta^2) + (x^2)^2 + (x^3)^2 \\
& = -(x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2
\end{aligned}$$

$$\text{as } \left(\gamma^2 = \frac{1}{1 - \beta^2} \right)$$

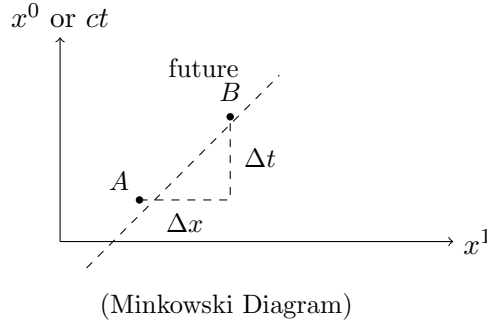
$$-(x^0)^2 + d^2 = I = \text{const}$$

$$\text{where, } d^2 = (x^1)^2 + (x^2)^2 + (x^3)^2$$

Proper time :-

Event A : particle at A

Event B : particle reach B



let us see space time interval b/w these two events $A(x_A^{(0)}, x_A^{(1)})$ and $B(x_B^{(0)}, x_B^{(1)})$

$$(\Delta s)^2 = -(\Delta x^0)^2 + (\Delta x^1)^2 = -(\Delta \bar{x}^0)^2 + (\Delta \bar{x}^1)^2$$

↘ In frame of particle ($\Delta \bar{x}^1 = 0$)

$$-c^2 \Delta t^2 + \Delta x^2 = -c^2 (\Delta \tau)^2 + 0$$

$$\Rightarrow c^2 (\Delta \tau)^2 = c^2 (\Delta t)^2 - (\Delta x)^2$$

$$\text{So time in particle's frame } (\Delta \tau) = \Delta t \sqrt{1 - \frac{(\frac{\Delta x}{\Delta t})^2}{c^2}}$$

$$\boxed{\Delta \tau = \Delta t \sqrt{1 - \frac{v^2}{c^2}}} \quad \text{or} \quad (d\tau = dt \sqrt{1 - v^2/c^2})$$

Four velocity :-

Imagine a particle moving in space-time.

$$\text{We define four velocity } (V \text{ or } \eta) = \frac{\Delta \vec{x}}{\Delta \tau} = \frac{d\vec{x}}{d\tau}$$

$$\Delta \vec{x} = \vec{x}_2 - \vec{x}_1$$

$\Delta \tau$ = time elapsed in particles frame

$$V = \frac{d\bar{x}}{d\tau} \quad \text{or} \quad \eta^\mu \text{ or } V^\mu = \frac{dx^\mu}{d\tau}$$

$\searrow$ has four components ($\mu = 0, 1, 2, 3$)

$$\eta = \left(\frac{dx^0}{d\tau}, \frac{dx^1}{d\tau}, \frac{dx^2}{d\tau}, \frac{dx^3}{d\tau} \right) = \frac{1}{\sqrt{1-\beta^2}} \left(\frac{dx^0}{dt}, \frac{dx^1}{dt}, \frac{dx^2}{dt}, \frac{dx^3}{dt} \right)$$

$$\eta = \frac{1}{\sqrt{1-\beta^2}} \left(c \frac{dt}{dt}, \frac{d}{dt}(\vec{r}) \right) = \gamma(c, \vec{v})$$

Four Momentum :-

$$\vec{p} = m_0 \vec{V} \quad \text{or} \quad m_0 \eta^\mu$$

$$p^\mu = m_0 \eta^\mu = \left(\frac{m_0 c}{\sqrt{1-\beta^2}}, \frac{m_0 \vec{V}}{\sqrt{1-\beta^2}} \right) = \gamma(m_0 c, m_0 \vec{V})$$

$$\left\{ p^\mu = m_0 \frac{dx^\mu}{d\tau} = m_0 \eta^\mu \right\} \quad \text{I am writing again to get used to the notation.}$$

$$\vec{P} = (p^0, p^1, p^2, p^3) \quad p^\mu = m_0 \eta^\mu$$

Using $(d\tau = dt \sqrt{1-u^2/c^2})$ u = speed of particle

$$\text{So, } p^0 = m_0 \eta^0 = m_0 \frac{dx^0}{d\tau} = \frac{m_0}{\sqrt{1-v^2/c^2}} \frac{dx^0}{dt} = \frac{m_0 c}{\sqrt{1-v^2/c^2}}$$

$$p^1 = m_0 \eta^1 = m_0 \frac{dx^1}{d\tau} = \frac{m_0 v_x}{\sqrt{1-v^2/c^2}}$$

$$p^2 = m_0 \eta^2 = m_0 \frac{dx^2}{d\tau} = \frac{m_0 v_y}{\sqrt{1-v^2/c^2}}$$

$$p^3 = m_0 \frac{dx^3}{d\tau} = \frac{m_0 v_z}{\sqrt{1-v^2/c^2}}$$

Lets know these components of four-momentum.

$$\begin{aligned} p^0 &= \frac{m_0 c}{\sqrt{1-v^2/c^2}} = m_0 c \left(1 - \frac{v^2}{c^2} \right)^{-1/2} \\ &= m_0 c \left(1 + \frac{v^2}{2c^2} + \frac{3}{8} \frac{v^4}{c^4} + \dots \right) \end{aligned} \quad \begin{cases} \frac{v}{c} < 1 \\ (1+x)^n = 1 + nx + \frac{n(n-1)}{2} x^2 + \dots \end{cases}$$

$$p^0 = m_0 c + \frac{1}{2} m_0 v^2 \frac{1}{c} + \frac{3}{8} m_0 \frac{v^4}{c^3} + \dots$$

$$E = c p^0 = \underbrace{m_0 c^2}_{\text{Rest mass energy}} + \underbrace{\frac{1}{2} m_0 v^2 + \frac{3}{8} m_0 \frac{v^4}{c^2} + \dots}_{\text{K.E}}$$

So physical Significance of 0th component is

$$p^0 = \frac{E}{c}$$

$$E = \text{K.E} + \text{Rest energy}$$

$$K = E - m_0 c^2$$

$$p^1 = \frac{m_0 v_x}{\sqrt{1 - v^2/c^2}} = p_x$$

$$p^2 = \frac{m_0 v_y}{\sqrt{1 - v^2/c^2}} = p_y$$

$$p^3 = \frac{m_0 v_z}{\sqrt{1 - v^2/c^2}} = p_z$$

$\vec{p}$ = relativistic momentum

$$\vec{p} = \frac{m_0 \vec{V}}{\sqrt{1 - v^2/c^2}} = \gamma m_0 \vec{V}$$

lets make our day